

5.2 Sigma Notation and limits of Finite Sums

In this section we introduce a more convenient notation for working with sums that have a large number of terms. After describing this notation and its properties, we consider what happens as the number of terms approaches infinity.

① Finite Sums and Sigma Notation

Sigma notation enables us to write a sum with many terms in the compact form

$$\sum_{k=1}^n a_k = a_1 + a_2 + a_3 + \cdots + a_n.$$

The Greek letter Σ stands for “sum”. The **index of summation** k tells us where the sum begins (at the number below the Σ symbol) and where it ends (at the number above Σ).

The diagram shows the sigma notation symbol $\sum_{k=1}^n a_k$. Four annotations with arrows point to different parts of the symbol:

- An arrow points to the n above the sigma, with the text "The index k ends at $k = n$."
- An arrow points to the $k = 1$ below the sigma, with the text "The index k starts at $k = 1$."
- An arrow points to the a_k to the right of the sigma, with the text " a_k is a formula for the k th term."
- An arrow points to the sigma symbol itself, with the text "The summation symbol (Greek letter sigma)".

Thus, we can write the squares of the numbers 1 through 11 as

$$1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + 9^2 + 10^2 + 11^2 = \sum_{k=1}^{11} k^2.$$

and the sum of $f(i)$ for integers i from 1 to 100 as

$$f(1) + f(2) + f(3) + \cdots + f(100) = \sum_{i=1}^{100} f(i).$$

◆ The starting index does not have to be 1; it can be any integer.

Example 1 Express the following sums in sigma notation.

(a) $1 + 2 + 3 + 4 + 5$;

(b) $(-1)^1 1 + (-1)^2 2 + (-1)^3 3;$

(c) $\frac{1}{1+1} + \frac{2}{2+1};$

(d) $\frac{4^2}{4-1} + \frac{5^2}{5-1}.$

Example 2 Express the sum $1 + 3 + 5 + 7 + 9$ in sigma notation.

② Rules for Finite Sums

Algebra Rules for Finite Sums

1. *Sum Rule:* $\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k$
2. *Difference Rule:* $\sum_{k=1}^n (a_k - b_k) = \sum_{k=1}^n a_k - \sum_{k=1}^n b_k$
3. *Constant Multiple Rule:* $\sum_{k=1}^n c a_k = c \cdot \sum_{k=1}^n a_k$ (Any number c)
4. *Constant Value Rule:* $\sum_{k=1}^n c = n \cdot c$ (Any number c)

Example 3 We demonstrate the use of the algebra rules.

(a) $\sum_{k=1}^n (3k - k^2);$ (b) $\sum_{k=1}^n (-a_k);$ (c) $\sum_{k=1}^3 (k + 4);$ (d) $\sum_{k=1}^n \frac{1}{n}.$

Example 4 Show that the sum of the first n integers is

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

Note 1 The first n squares: $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$

③ limits of Finite Sums

Example 5 Find $\lim_{n \rightarrow \infty} \sum_{k=1}^n f\left(\frac{k}{n}\right) \frac{1}{n}$, where $f\left(\frac{k}{n}\right) = 1 - \left(\frac{k}{n}\right)^2$.

Example 6 Find $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k(k+1)}$.

④ Riemann Sums

We begin with an arbitrary bounded function f defined on a closed interval $[a, b]$.

We choose $n - 1$ points $\{x_1, x_2, x_3, \dots, x_{n-1}\}$ between a and b that are in increasing order, so that

$$a < x_1 < x_2 < x_3 < \dots < x_{n-1} < b.$$

Let $x_0 = a$ and $x_n = b$, so that

$$a = x_0 < x_1 < x_2 < x_3 < \dots < x_{n-1} < x_n = b.$$

The set of all of these points,

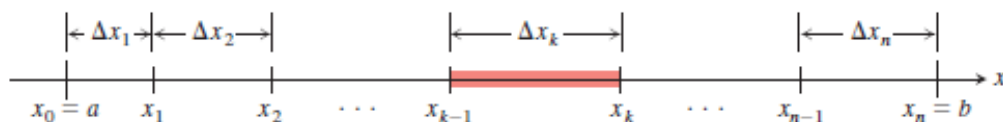
$$P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\},$$

is called a **partition** of $[a, b]$.

The partition P divides $[a, b]$ into the n closed subintervals

$$[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n].$$

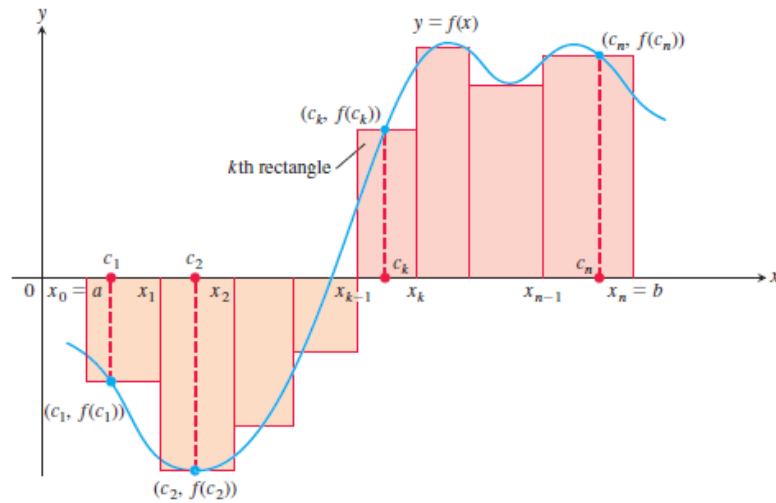
The width of the first subinterval $[x_0, x_1]$ is denoted Δx_1 , the width of the second $[x_1, x_2]$ is denoted Δx_2 , and the width of the k th subinterval is $\Delta x_k = x_k - x_{k-1}$.



We define **the norm of a partition P** , written $\|P\|$, to be the largest of all the subinterval widths.

In each subinterval we select some point. The point chosen in the k th subinterval

$[x_{k-1}, x_k]$ is called c_k .



On each subinterval we form the product $f(c_k)\Delta x_k$. This product is positive, negative, or zero, depending on the sign of $f(c_k)$. When $f(c_k) > 0$, the product $f(c_k)\Delta x_k$ is the area of a rectangle with height $f(c_k)$ and width Δx_k . When $f(c_k) < 0$, the product $f(c_k)\Delta x_k$ is a negative number, the negative of the area of a rectangle of width Δx_k that drops from the x -axis to the negative number $f(c_k)$.

Finally, we sum all these products to get

$$S_P = \sum_{k=1}^n f(c_k) \Delta x_k.$$

The sum S_P is called a **Riemann sum for f on the interval $[a, b]$** . There are many such sums, depending on the partition P we choose, and the choices of the points c_k in the subinterval.